Narendra Vanapalli

AI/ML Engineer | Full-Stack AI Product Engineer

+91 9493990003 | sivanarendravanapalli@gmail.com | LinkedIn | GitHub | Portfolio

Professional Summary

AI/ML Engineer with 2+ years of experience designing, developing, and deploying enterprise-grade Generative AI, agentic AI, and full-stack applications. Skilled in multi-agent orchestration, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), LLM evaluation, intelligent document processing, and AI guardrails. Proficient in Python, Java, Spring Boot, FastAPI, React.js, AWS, and Google Cloud, with strong expertise in microservices, Docker, Kubernetes, and CI/CD. Experienced in building scalable AI systems, integrating LLMs with enterprise data sources, and implementing evaluation frameworks to improve accuracy, reliability, and explainability. Adept at translating complex business requirements into secure, cloud-native solutions that deliver measurable business value.

Technical Skills

Languages: Python, Java, SQL, JavaScript
AI & ML: Generative AI, LLMs, RAG, embeddings, vector search, AI evaluation, guardrails, MLOps, LLMOps
Models & Frameworks: GPT, Claude, Gemini, Amazon Bedrock, Vertex AI, OpenAI APIs, LangChain, LangGraph, Strands Agents, Google ADK, OpenAI Agents SDK, MCP, LiteLLM
Development: FastAPI, Spring Boot, Spring WebFlux, React, Redux, REST APIs, microservices
Data & Cloud: PostgreSQL, Redshift, SQL Server, MongoDB, Redis, pgvector, Pinecone, Qdrant, ChromaDB, AWS, Google Cloud, Amazon ECS, Amazon EKS, Docker, Kubernetes
Quality & Delivery: RAGAS, Langfuse, pytest, JUnit, Playwright, Git, GitHub, GitHub Actions, CI/CD, Jira

Professional Experience

Endava Solutions India Private Limited **Aug 2024 – Present | Bengaluru, India**
Machine Learning Engineer | Associate Developer → Junior ML Engineer → ML Engineer

- Contributed to **6 enterprise AI and full-stack initiatives** across healthcare, analytics, grant compliance, and workforce intelligence, spanning solution design, development, testing, and deployment.
- Engineered **Generative AI and agentic AI solutions** using RAG, multi-agent orchestration, MCP integrations, and LLM-powered workflows for document intelligence, natural-language analytics, and automated decision support.
- Developed **scalable microservices and full-stack applications** using Python, FastAPI, Java Spring Boot/WebFlux, React.js, and PostgreSQL, incorporating asynchronous processing, API integrations, and performance optimization.
- Implemented **AI evaluation and reliability frameworks** using gold-standard datasets, retrieval metrics, citation validation, guardrails, and human-in-the-loop review to improve output quality and explainability.
- Supported **AWS and Google Cloud deployments** using Docker, Kubernetes, CI/CD, and infrastructure automation to deliver scalable, cloud-native applications.
- Collaborated with **engineering, QA, product, and business stakeholders** through Agile ceremonies, technical reviews, debugging, demonstrations, and release coordination to deliver solutions aligned with business requirements.

GalaxE Solutions, Inc. **Feb 2024 – Apr 2024 | Bengaluru, India**
Software Engineering Intern

- Independently engineered a **full-stack e-commerce platform** using **Java Spring Boot, React.js, and SQL databases**, implementing **6+ business workflows** spanning product discovery, inventory management, checkout, order fulfillment, payments, and delivery tracking.
- Developed **RESTful APIs and modular backend services** with **JWT authentication, role-based access control, and WebSocket-based real-time updates**, enabling secure transactions and live order-status tracking.
- Optimized application responsiveness through **multithreading, asynchronous processing, and database query optimization**, while implementing **JUnit and Mockito tests** to improve backend reliability and maintainability.

Selected Projects

ClaimFlow – AI-Powered Healthcare Claims Management Platform *Enterprise Project*

Java, Spring Boot, Python, FastAPI, React.js, RAG, gRPC, AWS, Docker, Kubernetes

- Engineered an **AI-powered healthcare claims management platform** using Spring Boot, FastAPI, and React.js, streamlining claim submission, policy verification, eligibility assessment, and end-to-end workflow tracking.
- Designed **Java and Python microservices** with RESTful APIs and **gRPC-based communication**, separating core claims processing from AI-driven validation and document intelligence workflows.
- Developed an **OCR and RAG-powered document intelligence pipeline**, achieving **95% critical-field extraction accuracy and 0.90 claim-validation F1** through semantic policy retrieval, LLM-based assessment, and evidence-grounded validation.

- Implemented **event-driven processing** using Amazon SNS, SQS, and AWS Lambda to orchestrate asynchronous claim validation and retryable background workflows, achieving **p95 automated assessment turnaround below 2 minutes**.
- Containerized and deployed microservices using **Docker, Kubernetes, and Amazon EKS**, integrating Amazon API Gateway, S3, and DynamoDB for API access, secure document storage, and workflow-state management.

Grant Compliance Review Platform – AI-Powered Compliance Intelligence

Ford Foundation

Python, FastAPI, React.js, Vertex AI, Gemini, RAG, Firestore, Google Cloud, Docker, CI/CD

- Engineered an **AI-powered grant compliance review platform** using FastAPI, React.js, and **Google Cloud Vertex AI**, automating customer-specific compliance assessments and generating **evidence-backed, reviewer-ready PDF reports**.
- Architected a **dynamic, multi-format RAG pipeline** with **on-demand document ingestion, semantic chunking, and contextual retrieval** across PDF, Word, and Excel files, adapting to heterogeneous customer submissions and compliance requirements.
- Integrated **Gemini models on Vertex AI** for **criterion-level compliance assessment**, generating structured findings with verifiable source citations, evidence validation, and LLM guardrails to support explainable reviews.
- Developed a **gold-standard evaluation framework**, achieving **0.88 criterion-level Macro F1 and 95% citation precision**, while maintaining a **false-compliant rate below 5%** across expert-reviewed compliance assessments.
- Deployed **containerized services on Google Cloud** with automated **CI/CD pipelines, IAM, Secret Manager, and Workload Identity Federation**, enabling secure, repeatable application releases.

ProductMatchAI – AI-Powered Medical Device Competitor Intelligence

J&J

Python, FastAPI, React.js, Amazon Bedrock, Amazon EKS, PostgreSQL, pgvector, Docker, Hybrid Search

- Spearheaded the **end-to-end design and development** of an AI-powered medical device competitor intelligence platform as the **sole engineer**, enabling individual and bulk competitor analysis against a searchable catalog of **6M+ medical device records**.
- Architected a **multi-stage AI matching pipeline** combining **hybrid keyword-vector retrieval, cross-encoder reranking, and LLM-based relevance assessment** to deliver **ranked top-10 competitor recommendations** for analyst validation.
- Integrated **Amazon Bedrock-hosted LLMs** to interpret complex medical device descriptions and technical specifications, generating **context-aware comparisons, explainable rankings, and confidence-based recommendations** to support informed analyst decisions.
- Developed an **expert-labeled evaluation framework**, achieving **92% Recall@200, 95% Hit Rate@10, and 0.82 nDCG@10**, validating candidate retrieval, matching effectiveness, and final ranking quality.
- Containerized and deployed the platform using **Docker, Kubernetes, and Amazon EKS**, supporting scalable, cloud-native workloads; reduced manual competitor-review effort by approximately **60%** through AI-assisted matching and ranked recommendations.

Endava Talent Sphere – Workforce Intelligence Platform

Internal Product

Java, Spring WebFlux, React.js, PostgreSQL, Amazon EKS, Docker, Kubernetes, Caffeine, CI/CD

- Developed core modules of an **enterprise workforce intelligence platform** using **Java Spring WebFlux, React.js, and PostgreSQL**, supporting employee profiles, organizational hierarchies, skill inventories, and project staffing workflows for **5,000+ internal users**.
- Designed and implemented **reactive Spring WebFlux microservices and non-blocking REST APIs**, enabling skill-based talent discovery, resource allocation tracking, and workforce planning workflows.
- Reduced API response latency by **87% (3,000 ms to 400 ms)** through PostgreSQL query tuning, indexing, reactive connection management, and **Caffeine caching**.
- Implemented **SSO, role-based access control (RBAC), Server-Sent Events (SSE), and asynchronous email notifications**, supporting secure workflows and real-time updates across employee, manager, and administrator interfaces.
- Contributed to **Docker-based microservice deployments on Amazon EKS**, integrating Kubernetes configurations, CI/CD automation, testing, and health checks to support reliable application releases.

Recognitions & Certifications

- **InSync Regional Hackathon 2026:** Led Team Orchestrators in building Dava Compass, an AI-assisted workforce planning platform. The team reached the **top 10** and received hackathon recognition.
- **Endava AI Days 2025:** Secured **second place** in the Speed Coding Challenge for AI Evaluator, a Generative AI platform for rubric-based code assessment.
- **Databricks Certified Generative AI Engineer Associate**
- **AWS Certified AI Practitioner**.
- Completed courses through DeepLearning.AI and Udemy covering RAG, agentic AI, LLM engineering, and responsible AI guardrails.
- Recognized as the **top performer among more than 80 trainees** in the GalaxE Solutions Software Engineering training program.

Education

University College of Engineering, Kakinada (JNTUK) — **2020 – 2024**

Bachelor of Technology in Electronics and Communication Engineering — *Kakinada, India*